

Multi-Head LUT Detectors: Anchor-Pairs, Hyperplane, and LIF

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Abstract

The Multi-Head LUT (MHL) family builds a differentiable function from a bank of small learned lookup tables whose row addresses are produced by a bank of *detectors* applied to the input. This note describes the shared addressing–lookup–sum skeleton and three choices of detector: a fixed coordinate comparison (anchor-pairs), a learned affine half-space (hyperplane), and a spiking leaky integrate-and-fire (LIF) neuron. The first two discretize the input *geometrically*; the third discretizes it *temporally*, encoding the input as spike latencies and reading out *when* a neuron first fires. It is a mechanism/derivation note only — no experimental results are reported.

1 The shared skeleton

Let the input be $\mathbf{x} \in \mathbb{R}^{d_{\text{in}}}$ (batch size B). A module has H output heads and P tables per head, hence $T = H P$ lookup tables. Each table t carries D *detectors*; each detector emits a single integer *digit*; the digits index a learned table of cells. The output is a tensor of shape (B, H, d_{out}) .

Every MHL variant is the same four steps.

(1) Per-detector discretization. Detector d of table t maps the input to a digit

$$g_{t,d}(\mathbf{x}) \in \{0, 1, \dots, r_d - 1\}, \quad (1)$$

where r_d is that detector’s *radix* (the number of symbols it may emit).

(2) Mixed-radix cell address. The D digits of a table are packed, row-major, into one cell index

$$\text{idx}_t = \sum_{d=0}^{D-1} g_{t,d} \text{radix}_d, \quad \text{radix}_d = \prod_{d' > d} r_{d'}, \quad (2)$$

so the table has $\prod_d r_d$ rows (cells). When every $r_d = 2$ this is simply reading the digits as a binary number, most-significant bit first.

(3) Table read. Each table is a learned tensor $\text{Table}_t \in \mathbb{R}^{\text{cells} \times d_{\text{out}}}$, and the addressed row is

$$\text{row}_t = \text{Table}_t[\text{idx}_t] \in \mathbb{R}^{d_{\text{out}}}. \quad (3)$$

(4) **Head sum** $\rightarrow$ **output**. The P tables of head h are summed:

$$\mathbf{y}_h = \sum_{p=1}^P \text{row}_{(h,p)} \in \mathbb{R}^{d_{\text{out}}}, \quad \mathbf{y} = (\mathbf{y}_1, \dots, \mathbf{y}_H) \in \mathbb{R}^{H \times d_{\text{out}}}. \quad (4)$$

The tables are the only memory; the detectors are the only thing that reads structure out of $\mathbf{x}$. The three variants below differ *only* in the choice of detector. Two of them (anchor-pairs, hyperplane) are *geometric*: a digit says which side of a boundary $\mathbf{x}$ lies on, with radix $r_d = 2$. The third (LIF) is *temporal*: a digit says *when* a spiking neuron first fires, with radix $r_d = M$ time buckets.

2 Anchor-pairs detectors (radix 2, coordinate comparison)

The simplest detector is a fixed ordered pair of input coordinates (a_d, b_d) . Its digit is the sign of their difference,

$$g_{t,d}(\mathbf{x}) = \mathbf{1}[x_{a_{t,d}} - x_{b_{t,d}} > \varepsilon] \in \{0, 1\}, \quad \varepsilon \geq 0, \quad (5)$$

where ε is a comparison margin. Each table has D such pairs; the D bits form a binary index into a table of 2^D rows via (2). The pairs are chosen once (balanced coverage over the input coordinates) and held fixed; only the tables are learned in the base construction.

Because the boundary $x_a = x_b$ is a hard threshold, the digit is piecewise constant in $\mathbf{x}$ and the raw map is non-differentiable. The module additionally exposes, per table, *alternative* indices obtained by flipping the bits with the smallest margins $|x_{a_d} - x_{b_d}|$, together with a per-bit *uncertainty* weight (for example inverse- L_1 , $\propto 1/|x_{a_d} - x_{b_d}|$). These quantify which detector is closest to flipping and are what a surrogate backward pass uses to route gradients to $\mathbf{x}$ and to neighbouring cells. Geometrically each detector is the axis-pair hyperplane $x_a - x_b = 0$.

3 Hyperplane detectors (radix 2, learned affine)

This replaces the fixed coordinate pair by a learned affine half-space test. Detector d of table t owns a weight row $\mathbf{w}_{t,d} \in \mathbb{R}^{d_{\text{in}}}$ and a bias $b_{t,d} \in \mathbb{R}$, and emits

$$g_{t,d}(\mathbf{x}) = \mathbf{1}[\langle \mathbf{w}_{t,d}, \mathbf{x} \rangle + b_{t,d} > 0] \in \{0, 1\}. \quad (6)$$

Again D bits pack most-significant-first into a 2^D -row table index, and the P tables of a head are summed. This strictly generalizes Section 2: initializing $\mathbf{w}_{t,d} = \mathbf{e}_a - \mathbf{e}_b$ (a difference of standard basis vectors) and $b_{t,d} = 0$ gives $\langle \mathbf{w}_{t,d}, \mathbf{x} \rangle = x_a - x_b$, exactly the anchor-pairs bit (5). The added freedom is that each detector's boundary is now an arbitrary oriented hyperplane in input space rather than an axis-pair diagonal.

Training is end-to-end through the hard $\arg \max$ /gather via a full- K softmax surrogate: the forward value is the hard table row, while the backward pass replaces each hard sign by a soft membership $\sigma((\langle \mathbf{w}, \mathbf{x} \rangle + b)/\tau)$ and forms a soft distribution over the 2^D rows, so gradients reach $\mathbf{w}$, b , the table weights, and $\mathbf{x}$. Two temperatures (a per-bit score temperature and a row-select temperature) control the sharpness of that surrogate.

4 LIF detectors (radix M , temporal / first-spike)

Here a detector is a spiking leaky integrate-and-fire (LIF) neuron, and its digit is *when* it first fires, discretized into M time buckets. A table carries D such neurons, giving M^D cells. The case $D = 1$

is the “bucket” variant (one neuron, M time buckets $\rightarrow M$ cells); $D > 1$ is the “product” variant (a mixed-radix address over several neurons $\rightarrow M^D$ cells).

(a) Latency coding. The real input is first turned into input *spike times*: each coordinate x_i fires once, earlier for larger x_i , through a fixed affine-then-clamp map onto a time window $[0, t_w]$,

$$\ell_i = \text{clamp}\left(t_w\left(\frac{1}{2} - \frac{3x_i}{32}\right), 0, t_w\right). \quad (7)$$

At $t_w = 32$ this is $\ell_i = \text{clamp}(16 - 3x_i, 0, 32)$: $x_i = 0$ fires at the window centre, the map saturates at $x_i = \pm 16/3 \approx \pm 5.33$, and it is monotonically *decreasing* (a larger input fires earlier). The map has no learned parameters and expects roughly unit-variance input.

(b) Synapses, delays, arrivals. Detector d has one synapse per input coordinate, with a bounded excitatory weight and a non-negative delay,

$$w_{d,i} = w_{\max} \sigma(w_{d,i}^{\text{raw}}) \in (0, w_{\max}), \quad \delta_{d,i} = \text{clamp}(\delta_{d,i}^{\text{raw}}, 0, t_w), \quad (8)$$

where σ is the logistic function. The arrival time of input i at neuron d is the input spike time plus the synaptic delay,

$$a_{d,i} = \ell_i + \delta_{d,i} \in [0, 2t_w]. \quad (9)$$

(c) LIF membrane and first spike. The neuron is a leaky integrator with time constant $\tau_d = \text{softplus}(\tau_d^{\text{raw}}) + 1$. Its membrane potential is the sum of exponentially decaying post-synaptic bumps from all arrivals up to time t ,

$$V_d(t) = \sum_{i: a_{d,i} \leq t} w_{d,i} \exp\left(-\frac{t - a_{d,i}}{\tau_d}\right). \quad (10)$$

Evaluated at the arrival instants (after sorting them in time), (10) is computed in a single pass by the factorization

$$V = \exp(-a/\tau) \odot \text{cumsum}(w \odot \exp(a/\tau)), \quad (11)$$

which is $O(N)$ in the number of synapses N . The neuron fires at its *first-spike time*, the earliest arrival at which the membrane crosses a fixed threshold $\theta = 1$,

$$t_d^* = \min\{a_{d,i} : V_d(a_{d,i}) \geq \theta\}, \quad t_d^* = t_w \text{ if it never crosses.} \quad (12)$$

A detector’s output is thus a single scalar *time* $t_d^* \in [0, t_w]$ — the temporal analogue of a hyperplane’s scalar score $\langle \mathbf{w}, \mathbf{x} \rangle + b$.

(d) Temporal bucketing $\rightarrow$ digit. Each detector owns $M - 1$ trainable, strictly increasing bucket boundaries, kept ordered by construction,

$$\beta_{d,k} = \beta_d^{\text{base}} + \sum_{j \leq k} \text{softplus}(\beta_{d,j}^{\text{raw}}), \quad \beta_{d,1} < \beta_{d,2} < \dots < \beta_{d,M-1}. \quad (13)$$

The digit counts how many boundaries the first-spike time has passed,

$$g_{t,d}(\mathbf{x}) = \#\{k : t_d^* \geq \beta_{d,k}\} = \sum_{k=1}^{M-1} \mathbf{1}[t_d^* \geq \beta_{d,k}] \in \{0, \dots, M-1\}. \quad (14)$$

This is a radix- M digit (versus the radix-2 bit of Sections 2–3). Steps (a)–(d) produce one digit per detector; the digits then run through the shared mixed-radix address, table read, and head sum of Section 1.

(e) **Differentiable training.** The hard path — the arg min first-spike (12) and the boundary count (14) — is piecewise constant. Training uses a straight-through estimator: the forward value is the hard cell read $\mathbf{y}_{\text{hard}}$ (which also carries the gradient to the selected table row), while a parallel soft readout $\mathbf{y}_{\text{addr}}$ supplies the address gradient, combined so the forward stays exactly hard,

$$\mathbf{y} = \mathbf{y}_{\text{hard}} + \mathbf{y}_{\text{addr}} - \text{sg}(\mathbf{y}_{\text{addr}}), \quad (15)$$

where $\text{sg}(\cdot)$ is the stop-gradient operator. The soft readout replaces the two hard steps by smooth surrogates: a soft first-spike time from per-arrival crossing (“escape”) probabilities $c_k = \sigma((V_k - \theta)/T_{\text{cross}})$ with survival $\prod_{j < k} (1 - c_j)$, giving an expected spike time; and a soft bucket partition over the M buckets from differences of $\sigma((t - \beta_{d,k})/T_{\text{bkt}})$. The soft distribution over cells is read against a *detached* copy of the table, so that (15) sends value and table-weight gradients through the hard read while the soft read sends address gradients to the weights, delays, τ , and boundaries. Two temperatures $T_{\text{cross}}, T_{\text{bkt}}$ set the softness (they may be frozen to 1). Inference uses the pure hard path with no soft computation.

5 How the three connect, and where LIF differs

Side by side, the variants are one backbone with three detector choices (Table 1); everything from the digit onward — the mixed-radix index (2), the table read, and the head sum — is identical.

Variant	Detector	Digit g_d	Radix r_d
Anchor-pairs	fixed coordinate pair	$\mathbf{1}[x_a - x_b > 0]$	2
Hyperplane	learned affine half-space	$\mathbf{1}[\langle \mathbf{w}, \mathbf{x} \rangle + b > 0]$	2
LIF	spiking LIF neuron, 1st spike	$\#\{k : t^* \geq \beta_k\}$	M

Table 1: The three detectors share the addressing and table-read backbone and differ only in how a digit is produced. A table then has r_d^D cells: 2^D for the geometric variants, M^D for LIF.

The geometric variants discretize the input *spatially*: a detector answers “which side of a boundary is $\mathbf{x}$ on?”, the boundary being a coordinate-pair diagonal (anchor-pairs) or a general learned hyperplane (hyperplane, which contains anchor-pairs as an initialization). Each detector yields one bit, and a table read is a lookup on a sign pattern.

The LIF variant discretizes the input *temporally*: $\mathbf{x}$ becomes a pattern of spike latencies (7), a LIF neuron integrates delayed, weighted, leaky charge (10), and the detector reports *when* it first fires (12), sorted into M learned time buckets (14). Two consequences distinguish it from the geometric detectors.

- **Richer per-detector code.** A geometric detector emits one bit; a LIF detector emits one of M symbols (a radix- M digit), so a single neuron already splits the input into M regions rather than 2.
- **Dynamics instead of a static boundary.** The “boundary” is no longer a fixed hyperplane but the neuron’s temporal dynamics — per-synapse delays, a leak time constant τ , and a threshold θ — which decide both *whether* and *when* it fires. In the degenerate limit of no leak, no delays, and a single readout instant, threshold crossing on the summed weighted latencies reduces to a linear threshold on the ℓ_i , i.e. an affine test on $\mathbf{x}$. The LIF detector therefore *contains* the hyperplane detector as a special case and adds temporal structure (delays, leak, multiple time buckets) that a single hyperplane cannot express.

In short: the same lookup-table backbone throughout, but the detector moves from *geometry* (which side of a plane) to *time* (when a spiking neuron fires), trading a binary spatial test for a multi-valued temporal one.